MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 05
Chapter 11: Areas Related to Circles

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

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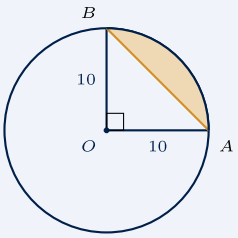
Q1. A chord of a circle of radius 10 cm subtends a right angle at the centre. The area of the minor segment is: (Use $\pi = 3.14$) [1]

(A) 28.5 cm^2 (B) 50 cm^2 (C) 78.5 cm^2 (D) 25 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Areas of Minor Sector and $\triangle OAB$ [0.5 Mark]

Radius $r = 10$ cm, $\theta = 90°$.



$$\text{Area of sector} = \frac{90°}{360°} \times 3.14 \times (10)^2 = \frac{314}{4} = 78.5 \text{ cm}^2$$

$$\text{Area of right } \triangle OAB = \frac{1}{2} \times 10 \times 10 = 50 \text{ cm}^2$$

Step 2: Area of Minor Segment [0.5 Mark]

$$\text{Area of segment} = 78.5 - 50 = 28.5 \text{ cm}^2$$

Correct Option: (A) 28.5 cm^2

Minor Segment Area = **28.5** cm^2

Q2. In a circle of radius 14 cm, a chord subtends an angle of 90° at the centre. The area of the major segment is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 560 cm^2 (B) 588 cm^2 (C) 616 cm^2 (D) 532 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Total Area and Minor Segment [0.5 Mark]

$$\text{Total Area of Circle} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$$

$$\text{Minor Sector Area} = \frac{1}{4} \times 616 = 154 \text{ cm}^2, \quad \text{Area}(\triangle OAB) = \frac{1}{2} \times 14 \times 14 = 98 \text{ cm}^2$$

$$\text{Minor Segment Area} = 154 - 98 = 56 \text{ cm}^2$$

Step 2: Area of Major Segment [0.5 Mark]

$$\text{Major Segment Area} = \text{Area of Circle} - \text{Minor Segment Area} = 616 - 56 = 560 \text{ cm}^2$$

Correct Option: (A) 560 cm^2

$$\boxed{\text{Major Segment Area} = \mathbf{560} \text{ cm}^2}$$

Q3. If the circumferences of two circles are in the ratio 4 : 9, then the ratio of their areas is: [1]

(A) 2 : 3 (B) 4 : 9 (C) 9 : 4 (D) 16 : 81

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Ratio of Radii [0.5 Mark]

$$\frac{C_1}{C_2} = \frac{2\pi r_1}{2\pi r_2} = \frac{r_1}{r_2} = \frac{4}{9}$$

Step 2: Ratio of Areas [0.5 Mark]

$$\frac{A_1}{A_2} = \frac{\pi r_1^2}{\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{4}{9}\right)^2 = \frac{16}{81} = 16 : 81$$

Correct Option: (D) 16 : 81

$$\boxed{\text{Ratio of Areas} = \mathbf{16 : 81}}$$

Q4. If the areas of two circles are in the ratio 25 : 36, then the ratio of their circumferences is: [1]

(A) 5 : 6 (B) 25 : 36 (C) 6 : 5 (D) 125 : 216

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Finding Radius Ratio from Area Ratio [0.5 Mark]

$$\frac{A_1}{A_2} = \left(\frac{r_1}{r_2}\right)^2 = \frac{25}{36} \implies \frac{r_1}{r_2} = \sqrt{\frac{25}{36}} = \frac{5}{6}$$

Step 2: Ratio of Circumferences [0.5 Mark]

$$\frac{C_1}{C_2} = \frac{2\pi r_1}{2\pi r_2} = \frac{r_1}{r_2} = \frac{5}{6} = 5 : 6$$

Correct Option: (A) 5 : 6

$$\boxed{\text{Ratio of Circumferences} = \mathbf{5 : 6}}$$

Q5. **Assertion (A):** The area of a minor segment of a circle is always strictly less than the area of the corresponding sector.
Reason (R): The area of a minor segment is obtained by subtracting the positive area of the corresponding triangle $\triangle OAB$ from the sector area. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Evaluating Reason (R) [0.5 Mark]

By definition, Area of minor segment = Area of sector OAB − Area of $\triangle OAB$. Since triangle area is strictly positive (Area > 0), Reason (R) is **True**.

Step 2: Evaluating Assertion (A) [0.5 Mark]

Subtracting a strictly positive value ensures Area(segment) $<$ Area(sector).
Thus Assertion (A) is **True**, and Reason (R) correctly explains (A).
Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Correct Option: (**A**)

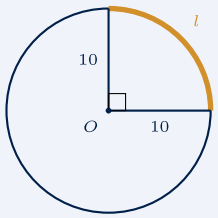
SECTION B: Very Short Answer Type

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Q6. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the length of the arc of the corresponding minor sector. (Use $\pi = 3.14$) [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Radius $r = 10$ cm, Central Angle $\theta = 90°$.



Step 1: Formula Declaration [1 Mark]

$$\text{Length of arc } l = \frac{\theta}{360°} \times 2\pi r = \frac{90°}{360°} \times 2\pi r = \frac{1}{4} \times 2\pi r = \frac{\pi r}{2}$$

Step 2: Substitution and Calculation [1 Mark]

$$l = \frac{3.14 \times 10}{2} = \frac{31.4}{2} = 15.7 \text{ cm}$$

Length of the arc l = **15.7** cm

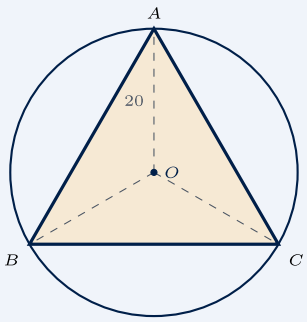
SECTION C: Short Answer Type

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Q7. Three students are seated at equal distances on the boundary of a circular park of radius 20 m. Find the length of each string connecting them and the area of the equilateral triangle formed. (Use $\sqrt{3} = 1.732$) **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let the three students be represented by vertices A, B, C of an equilateral triangle inscribed in a circular park with circumradius $R = 20$ m.



Step 1: Length of Side (String) a [1.5 Marks]

For an equilateral triangle of side a, circumradius $R = \frac{a}{\sqrt{3}}$:

$$a = R\sqrt{3} = 20\sqrt{3} \text{ m}$$

$$a = 20 \times 1.732 = 34.64 \text{ m}$$

Step 2: Area of Equilateral Triangle [1.5 Marks]

$$\text{Area} = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times (20\sqrt{3})^2 = \frac{\sqrt{3}}{4} \times 1200$$

$$\text{Area} = 300\sqrt{3} = 300 \times 1.732 = 519.6 \text{ m}^2$$

String Length $a = \mathbf{20\sqrt{3}}$ m $\approx \mathbf{34.64}$ m, Area of $\triangle = \mathbf{300\sqrt{3}}$ m^2 $= \mathbf{519.6}$ m^2

Q8. In a circle of radius 10 cm, a chord subtends a right angle at the centre. Find the area of:

(i) the minor segment, and

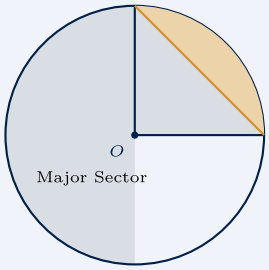
(ii) the major sector. (Use $\pi = 3.14$)

[3]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: Radius $r = 10$ cm, Central Angle $\theta = 90°$.

Part (i): Area of Minor Segment [1.5 Marks]



$$\text{Area of minor sector} = \frac{90°}{360°} \times 3.14 \times (10)^2 = \frac{314}{4} = 78.5 \text{ cm}^2$$

$$\text{Area of right } \triangle OAB = \frac{1}{2} \times 10 \times 10 = 50 \text{ cm}^2$$

$$\text{Area of minor segment} = 78.5 - 50 = 28.5 \text{ cm}^2$$

Part (ii): Area of Major Sector [1.5 Marks]

Central angle of major sector $\theta' = 360° - 90° = 270°$.

$$\text{Area of major sector} = \frac{270°}{360°} \times \pi r^2 = \frac{3}{4} \times 3.14 \times (10)^2$$

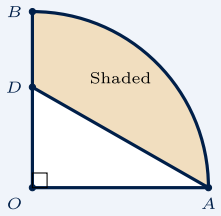
$$= \frac{3}{4} \times 314 = 3 \times 78.5 = 235.5 \text{ cm}^2$$

(i) Minor Segment Area $= \mathbf{28.5}$ cm^2, (ii) Major Sector Area $= \mathbf{235.5}$ cm^2

Q9. In the quadrant $OPBQ$ of a circle with centre O and radius 3.5 cm, a point D lies on OB such that $OD = 2$ cm. Find the area of the shaded region. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: Radius $r = 3.5$ cm $= \frac{7}{2}$ cm, $OD = 2$ cm.



Step 1: Area of Quadrant $OPBQ$ [1 Mark]

$$\text{Area of quadrant} = \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = \frac{77}{8} = 9.625 \text{ cm}^2$$

Step 2: Area of Right $\triangle AOD$ [1 Mark]

$$\text{Area}(\triangle AOD) = \frac{1}{2} \times \text{base}(OA) \times \text{height}(OD) = \frac{1}{2} \times 3.5 \times 2 = 3.5 \text{ cm}^2$$

Step 3: Area of Shaded Region [1 Mark]

$$\text{Shaded Area} = \text{Area of Quadrant} - \text{Area}(\triangle AOD)$$
$$= \frac{77}{8} - \frac{7}{2} = \frac{77 - 28}{8} = \frac{49}{8} = 6.125 \text{ cm}^2$$

$$\boxed{\text{Area of Shaded Region} = \mathbf{6.125} \text{ cm}^2 \quad \left(\text{or } \frac{\mathbf{49}}{\mathbf{8}} \text{ cm}^2\right)}$$

SECTION D: Case-Based / Applied Situation

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Q10. Case Study: Automobile Aerodynamic Dual Windshield Wipers

An automotive safety engineer designs a dual windshield wiper system for a passenger car. The system has two blades of length $r = 25$ cm each. Each wiper sweeps through a maximum operating angle of 115° without any overlapping region between the two blades.

(i) State the formula for the area swept by a wiper blade modeled as a sector of a circle. **[1]**

(ii) Calculate the area swept by a single wiper blade in one complete sweep. **[1]**

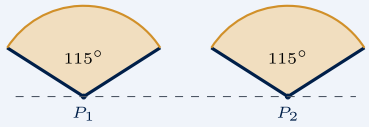
(iii) Calculate the total windshield surface area cleaned by both wipers simultaneously at each sweep. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[2]**

— OR —

Find the total arc distance traveled by the outer tip of both wiper blades. **[2]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given: Blade radius $r = 25$ cm, Sweep angle $\theta = 115°$.



Part (i): Formula for Swept Area [1 Mark]

$$\text{Area} = \frac{\theta}{360°} \times \pi r^2$$

Part (ii): Area Swept by Single Wiper Blade [1 Mark]

$$\text{Area}_1 = \frac{115°}{360°} \times \frac{22}{7} \times (25)^2 = \frac{23}{72} \times \frac{22}{7} \times 625 = \frac{158125}{252} \text{ cm}^2 \approx 627.48 \text{ cm}^2$$

Part (iii): Total Windshield Area Cleaned by Both Blades [2 Marks]

Since there is no overlap between the two blades:

$$\text{Total Area} = 2 \times \text{Area}_1 = 2 \times \frac{158125}{252} = \frac{158125}{126} \text{ cm}^2 \approx 1254.96 \text{ cm}^2$$

— OR (Alternative Choice) —

Total Arc Distance Traveled by Outer Tips: [2 Marks]

$$\text{Tip arc for 1 blade } l = \frac{115°}{360°} \times 2 \times \frac{22}{7} \times 25 = \frac{23}{72} \times \frac{1100}{7} = \frac{6325}{126} \text{ cm}$$

$$\text{Total Arc Distance for both blades} = 2 \times \frac{6325}{126} = \frac{6325}{63} \text{ cm} \approx 100.40 \text{ cm}$$

(i) $\frac{\theta}{360°}\pi r^2$, (ii) **627.48 cm²**, (iii) Total Cleaned Area = **1254.96 cm²** $\left(\text{or } \mathbf{\frac{158125}{126}} \text{ cm}^2\right)$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★